

Automated Diabetic Retinopathy Detection Using Deep Convolutional Neural Networks on Fundus Images

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ABSTRACT

Diabetic retinopathy (DR) affects 93 million people globally and is the leading cause of preventable blindness. Current screening programmes rely on trained graders, creating bottlenecks in high-prevalence regions. We present RetinaScan-Net, a deep CNN trained on 487,000 fundus images from 14 clinical sites across 6 countries. The model achieves 94.7% sensitivity and 96.2% specificity for referable DR, outperforming the average human grader (91.3% sensitivity, 93.8% specificity). Crucially, performance holds on images from low-resource settings with variable camera quality (sensitivity 92.1%). We deploy a lightweight variant (RetinaScan-Lite) that runs on commodity hardware, enabling point-of-care screening in settings without dedicated ophthalmic equipment.

METHODS

- Dataset: 487,000 fundus images; 112,000 with confirmed DR labels from three board-certified ophthalmologists
- Architecture: EfficientNet-B5 backbone with custom attention head for lesion localisation
- Training: 5-fold cross-validation; external validation on 3 held-out hospital datasets
- Grading categories: No DR, Mild, Moderate, Severe, Proliferative DR
- Hardware test: NVIDIA RTX 3060 (consumer GPU) — inference < 2 seconds per image

KEY FINDINGS

1. Model achieves near-expert sensitivity at 94.7% vs 91.3% for human graders
2. False negative rate 2.1% — lower than human graders in fatigue-simulated conditions
3. RetinaScan-Lite requires only 4GB RAM and runs on CPU in under 8 seconds
4. Inter-site variability handled by test-time augmentation and domain adaptation layer
5. Model identifies lesion sub-types (microaneurysms, haemorrhages, exudates) with 88% accuracy

LIMITATIONS

- Training data skewed toward European and East Asian populations; African dataset limited to 2 sites
- Model not validated on paediatric images
- Cannot distinguish between DR and other retinal pathologies (age-related macular degeneration, glaucoma)
- Regulatory approval required before clinical deployment in most jurisdictions

RELEVANCE TO ML IN CLINICAL DIAGNOSTICS: HIGH

This paper directly addresses a core challenge in clinical ML deployment: performance gap between research settings and real-world clinical environments. The multi-site validation and lightweight deployment are particularly relevant.